DO NOW — RECALL

Linked lists and pointers

Name: _____ Class: _____ Date: _____

1. Push 2, 5, 8 then pop. What remains bottom-to-top?

2. What is the first index of a C# array?

3. What is a base case?

4. Queue A,B,C: dequeue once and enqueue D. What remains?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2, 5. The top 8 is removed.

2. Model answer

0.

3. Model answer

The stopping condition in recursion.

4. Model answer

B,C,D front-to-rear.

Vocabulary — know these terms

node — A list element containing data and a link.

next pointer — Reference or index of the next node.

head — Reference to the first list node.

null sentinel — Marker such as −1 in an index implementation meaning no next node.

traversal — Visit nodes by following successive links.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm and explain

Then link node 2 to node 6, giving 2 → 6 → 8 → 11.

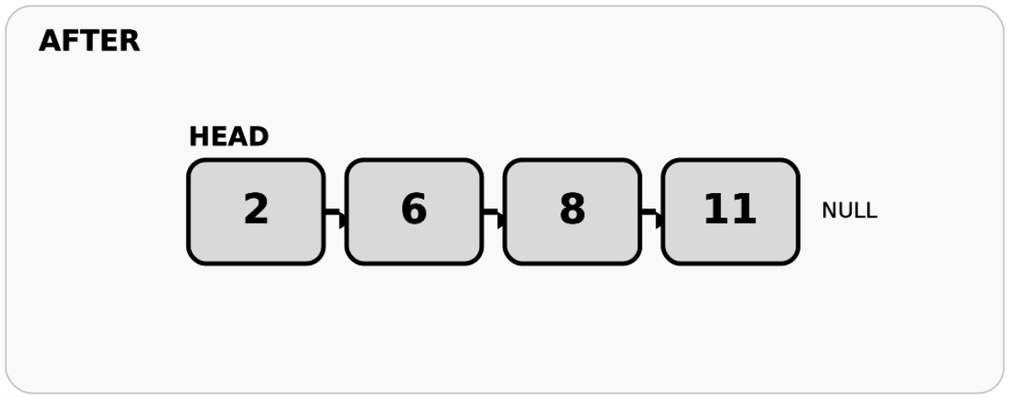
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 6 after 2 in 2 → 8 → 11 without losing 8.

First link new node 6 to node 8.



AFTER. 2, 6, 8, 11

NEW LEARNING

Read, annotate and explain

Follow references

head = 0 means the first node has value 10. next[0] initially points at node 1 (20).

Prepare the new node

The new node at index 3 stores 15 and next[3] = 1, preserving the rest of the list.

Change the existing link

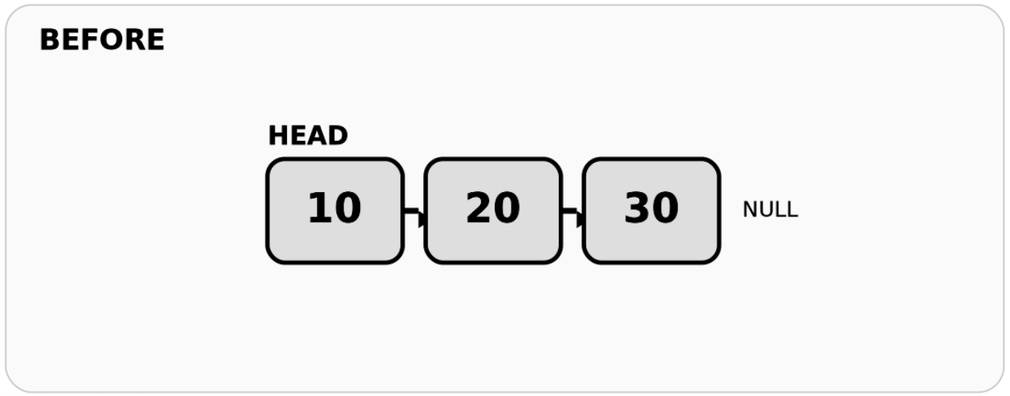
Set next[0] = 3, producing the chain 10 → 15 → 20 → 30.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

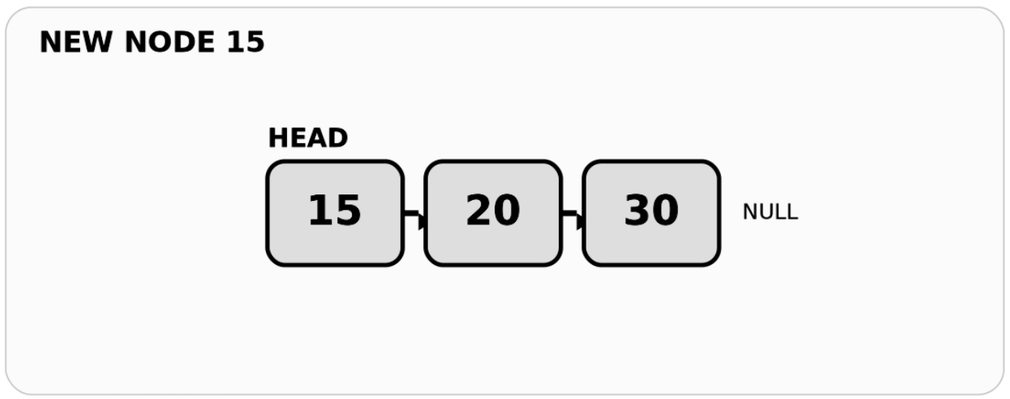
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BEFORE. 10, 20, 30

New Learning



NEW NODE 15. 15, 20, 30

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A singly linked list always supports direct O(1) access to its twentieth element.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

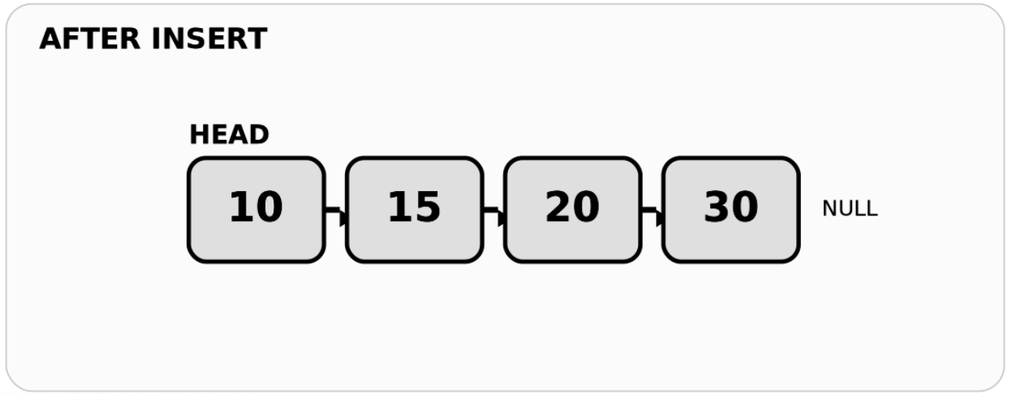
Question 3

Why does inserting into a linked list avoid shifting array elements but may still take O(n) time?

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



AFTER INSERT. 10, 15, 20, 30

I DO — WORKED EXAMPLE

Follow the worked method

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

Predict

Create node 7 and set its next link to the existing node 9.

Confirm and explain

Point node 4 at node 7. The chain becomes 4 → 7 → 9 → 12.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

In linked list 5 → 10 → 15, what follows node 10?

Question 2

A node with value 12 must be inserted between 7 and 20. State the two link changes.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict First link new node 6 to node 8.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

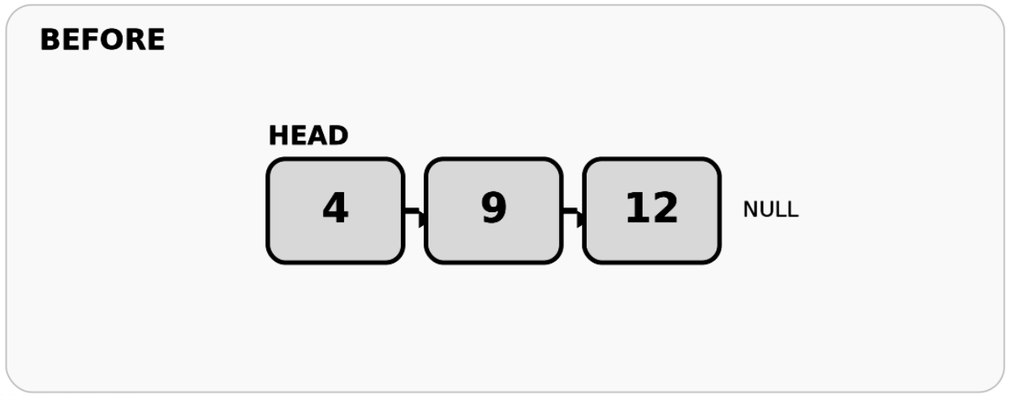
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

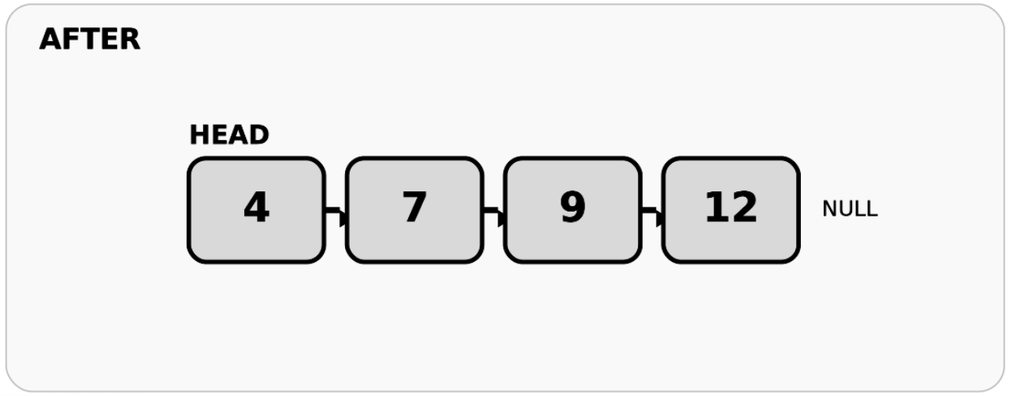
Create node 7 and set its next link to the existing node 9.



BEFORE. 4, 9, 12

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

Create node 7 and set its next link to the existing node 9.



AFTER. 4, 7, 9, 12

WE DO — GUIDED PRACTICE

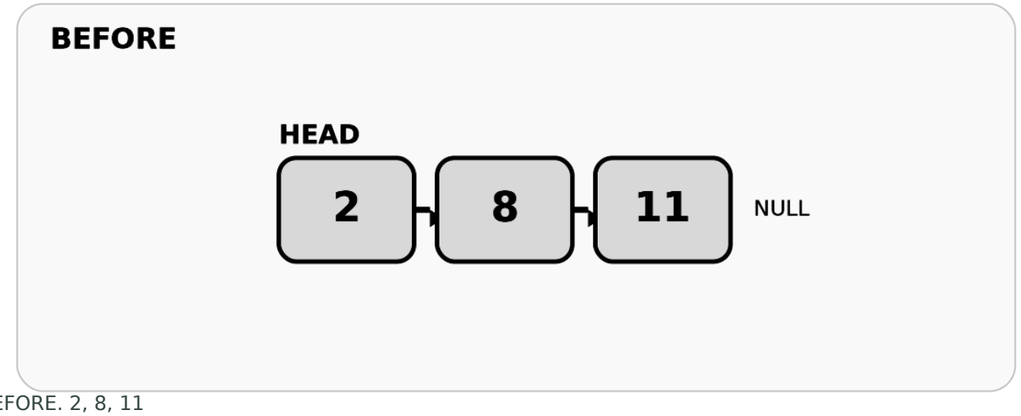
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 6 after 2 in 2 → 8 → 11 without losing 8.

First link new node 6 to node 8.



BEFORE. 2, 8, 11